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Assig : 04

Q1 (a) Context-Level DFD

Context-Level DFD (Level 0)

Context level mein poora system **ek single process** hota hai.

Main Process:

SafeHome Security System

External Entities:

1. **Home Owner**
2. **Police Department**
3. **Sensors**

Data Flow:

From	To	Data
Home Owner	System	Arm/Disarm Command
System	Home Owner	Alert Notification
Sensors	System	Intrusion Signal
System	Police	Emergency Alert

Context level mein sirf ek hi main circle hota hai (poora system).

Q1 (b) Define Attributes for the Class

Class Name: **SecuritySystem**

Attributes (Data Members)

1. systemID
2. location
3. status (Armed / Disarmed)
4. alarmState
5. sensorList
6. ownerName

Q1 (c) Four Operations (with Category)

1 Data Manipulation

addSensor()

System mein naya sensor add karega.

2 Inquiry

checkStatus()

System ka status return karega (Armed/Disarmed).

3 Monitoring

detectIntrusion()

Sensors ko monitor karega aur intrusion detect karega.

4 Computation

calculateResponseTime()

Alert bhejne ka response time calculate karega.

Q1 (d) Simple Class Diagram

Classes:

1. **SecuritySystem**
2. **Sensor**
3. **HomeOwner**
4. **Police**

Relationships:

- SecuritySystem **has many** Sensors ($1 \rightarrow *$)
- SecuritySystem **belongs to** one HomeOwner
- SecuritySystem **sends alert to** Police

Simple Text Diagram:

```
HomeOwner
|
| 1
SecuritySystem ----- Police
|
| 1..*
Sensor
```